

Math 432/532: Introduction to Topology II

Solutions to HW #3

For each of these problems, we will suppose that X is a topological space and $f : X \rightarrow X$ is a map such that $f(p) \neq p$ for all $p \in X$, but $f \circ f = \text{id}_X$. (In particular, $f(U) = f^{-1}(U)$ for any set U .) Define an equivalence relation on X by putting $p \sim f(p)$ for all $p \in X$ (so every equivalence class has exactly two elements). Let Y be the identification space $X/\sim$. Let $\pi : X \rightarrow Y$ be the identification map.

1. Assume that X is Hausdorff. The purpose of this problem is to prove that Y is Hausdorff, as well.

- (i) Show that, given any natural number n and any n -tuple of pairwise distinct points $p_1, \dots, p_n \in X$, we can find pairwise disjoint open sets $U_1, \dots, U_n$ with $p_i \in U_i$ for all i . (Note that the $n = 2$ case is the definition of Hausdorffness. For larger n , use induction. Note also that this part of the problem does not involve the map f or the space Y .)

Solution: Assume that we can do it with $n - 1$ points. Given $p_1, \dots, p_n$, we can use the inductive hypothesis to produce $U_1, \dots, U_{n-1}$. Since X is Hausdorff and $p_n \neq p_i$, we can find an open neighborhood V_i of p_n such that $U_i \cap V_i = \emptyset$. Then let $U_n = V_1 \cap \dots \cap V_{n-1}$.

- (ii) Let q_1 and q_2 be two distinct points in Y . Choose points $p_1, p_2 \in X$ with $\pi(p_i) = q_i$. In particular, this implies that the four points $p_1, f(p_1), p_2, f(p_2)$ are pairwise distinct. By part (i), we can find pairwise disjoint open sets U_1, U_2, V_1 , and V_2 with $p_1 \in U_1, p_2 \in U_2, f(p_1) \in V_1$, and $f(p_2) \in V_2$. Show that we can make these choices in such a way that $V_1 = f(U_1)$ and $V_2 = f(U_2)$.

Solution: First choose U_1, U_2, V_1 , and V_2 without this condition. Now define new open sets

$$U'_i := U_i \cap f(V_i) \quad \text{and} \quad V'_i = V_i \cap f(U_i).$$

These do the trick.

- (iii) Show that Y is Hausdorff.

Solution: Given two distinct points q_1 and q_2 in Y , choose p_1, p_2, U_1, U_2, V_1 , and V_2 as in part (ii). Let $W_i = \pi(U_i)$. Then $\pi^{-1}(W_i) = U_i \cup V_i$, so W_i is an open neighborhood of q_i . I claim that W_1 and W_2 are disjoint. Assume for the sake of contradiction that there exists a point $q \in W_1 \cap W_2$. Choose a point $p \in X$ with $\pi(p) = q$. Since $q \in W_1, p \in \pi^{-1}(W_1) = U_1 \cup V_1$. Similarly, $p \in U_2 \cup V_2$. This contradicts the fact that U_1, U_2, V_1 , and V_2 are pairwise disjoint.

2. Assume that X is a surface. The purpose of this problem is to prove that Y is a surface, as well. We have already established that, if X is Hausdorff, so is Y . So we just need to show that, if every point in X has an open neighborhood homeomorphic to an open subset of $\mathbb{R}^2$, then the same is true of Y .

- (i) Show that the identification map π is open.

Solution: Suppose $U \subset X$ is open. Then $\pi^{-1}(\pi(U)) = U \cup f(U)$, which is open, so $\pi(U)$ is open.

- (ii) Show that, if $U \subset X$ is an open set and $U \cap f(U) = \emptyset$, then $\pi : U \rightarrow \pi(U)$ is a homeomorphism.

Solution: It is obviously a surjection. If p_1 and p_2 are elements of U with $\pi(p_1) = \pi(p_2)$, then either $p_1 = p_2$ or $p_1 = f(p_2)$. But we can't have $p_1 = f(p_2)$ because $p_1 \in U$ and $f(p_2) \in f(U)$, so $p_1 = p_2$. This proves that $\pi : U \rightarrow \pi(U)$ is a bijection. It is open by part (i), so it is a homeomorphism.

- (iii) Show that, if $p \in X$, there exists an open neighborhood U of p such that U is homeomorphic to an open subset of $\mathbb{R}^2$ and $U \cap f(U) = \emptyset$.

Solution: First pick an open neighborhood A of p that is homeomorphic to an open subset of $\mathbb{R}^2$. Next, pick disjoint open neighborhoods B of p and C of $f(p)$. Then let $U = A \cap B \cap f(C)$. It's clearly an open neighborhood of p . Since A is homeomorphic to an open subset of $\mathbb{R}^2$ and U is an open subset of A , U is also homeomorphic to an open subset of $\mathbb{R}^2$. Finally, since $U \subset B$, $f(U) \subset C$, and B is disjoint from C , U is disjoint from $f(U)$.

(iv) Show that Y is a surface.

Solution: Let q be a point of Y , and choose a point $p \in X$ with $\pi(p) = q$. Choose an open neighborhood U of p as in part (iii). By parts (i) and (ii), $f(U)$ is an open neighborhood of q that is homeomorphic to U , and therefore to an open subset of $\mathbb{R}^2$.

For the remaining problems, suppose that $X = |K|$ is a combinatorial surface and $f = |\varphi|$ for some isomorphism $\varphi : K \rightarrow K$.

3. It's tempting to try to define a triangulation of Y whose simplices are in bijection with equivalence classes of simplices of K . Find an example that shows that such a simplicial complex might not exist! (If one subdivides K first, then it works, but you don't have to show that.)

Solution: Suppose that X is a torus and K looks a lot like Figure 6.4 in the textbook, but with a 4×4 grid rather than a 3×3 grid. Let f be given by shifting (cyclically) two units to the right. Then Y is also a torus, but the "triangulation" of Y that we try to define will look like a 2×4 grid, and that is not a simplicial complex. (For example, there is a pair of edges that share the same two endpoints.)

4. Suppose that K is equipped with an orientation. We say that φ is **orientation preserving** if it takes positively oriented triangles to positively oriented triangles. That is, if v_1, v_2, v_3 span a triangle in K and appear in positive cyclic order, then $\varphi(v_1), \varphi(v_2), \varphi(v_3)$ also appear in positive cyclic order. Give an example that is orientation preserving, and an example that is not.

Solution: The example in the solution to Problem 3 is orientation preserving. If we modify f by doing a shift *and* a flip over the horizontal axis, then it will be orientation reversing. In order for the flip to be simplicial, we should reverse the diagonals in the lower two rows (as discussed in class).

5. Assume that φ is orientation preserving. Also assume that the procedure described in Problem 3 actually works, so that we have a simplicial complex L and a homeomorphism $|L| \cong X/\sim$. Show that the orientation of K induces an orientation of L .

Solution: Suppose that w_0, w_1 , and w_2 are vertices of L that span a triangle. That triangle is the image of two triangles in K . One is spanned by vertices v_0, v_1 , and v_2 , with $\pi(v_i) = w_i$, and the other is spanned by the vertices $f(v_0), f(v_1)$, and $f(v_2)$. These two triangles are each oriented in K , and therefore induce orientations of their image in L . The fact that f is orientation preserving means that they induce the *same* orientation of their image in L .